```
Parameter Manifest — Hubbard-Holstein L=2

Model:
  - model_family: Hubbard-Holstein
  - problem: hh

Ansatz:
  - ansatz_type: hh_hva_ptw
  - vqe_method: hardcoded_hh_hva_physical_termwise_statevector
  - initial_state_source: vqe
  - branch_order: exact_gs_filtered, exact_paop, trotter_paop, exact_hva, trotter_hva

Drive Enabled: True

Branch Provenance:
  - paop_source: internal_adapt
  - paop_pool: uccsd_paop_lf_full
  - paop_depth: 30
  - hva_source: regular_vqe
  - hva_ansatz: hh_hva_ptw
  - hva_vqe_success: True
  - adapt_ref_source: vqe
  - A=0.5  omega=2.0
  - tbar=3.0  phi=0.0  t0=0.0
  - pattern=staggered  time_sampling=midpoint
  - exact_steps_multiplier=2

Core Physical Parameters:
  - t=1.0
  - U=4.0
  - dv=0.0

Hubbard-Holstein Parameters:
  - omega0=1.0
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  - g_ep=0.5
  - n_ph_max=1
  - boson_encoding=binary
  - nq_fermion=4
  - nq_phonon=2
  - nq_total=6

Trajectory Settings:
  - t_final=10.0
  - num_times=201
  - trotter_steps=64
  - suzuki_order=2
  - term_order=sorted
  - boundary=open
  - ordering=blocked
  - initial_state_source=vqe

Topline Results:
  - subspace_fidelity_at_t0: 0.9971190883301202
  - vqe_energy: 0.17229042845183531
  - exact_filtered_energy: 0.15866790412572704
  - qpe_energy_estimate: None
```
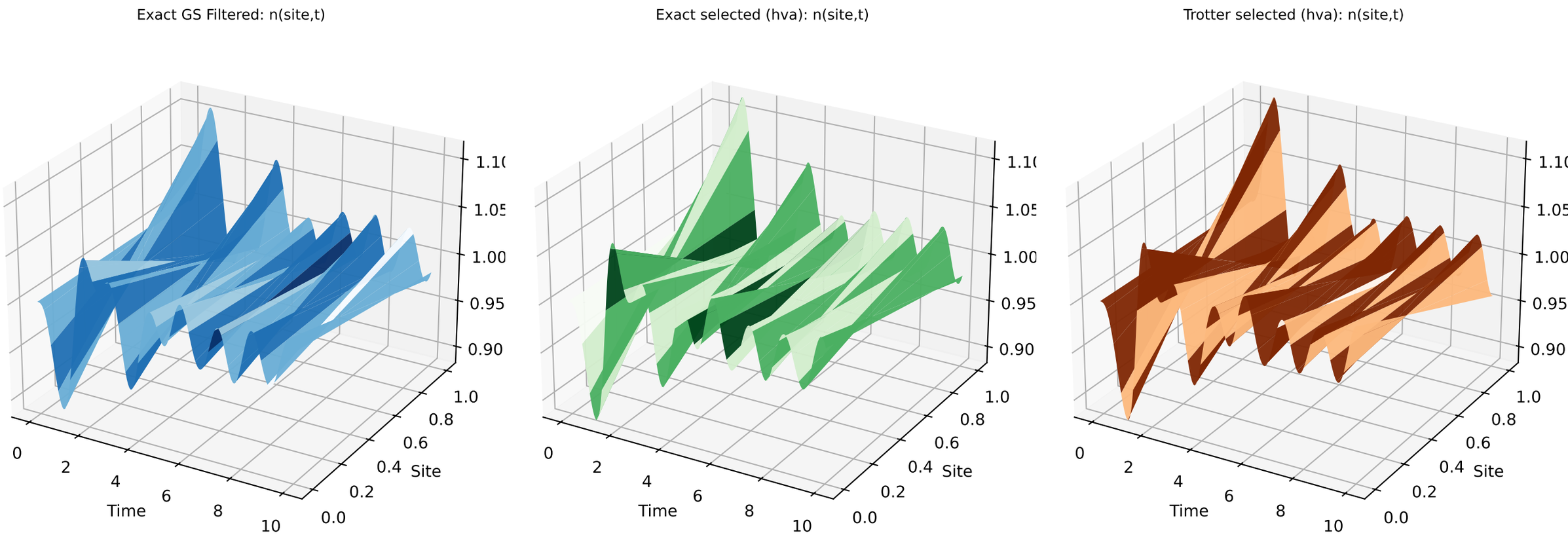
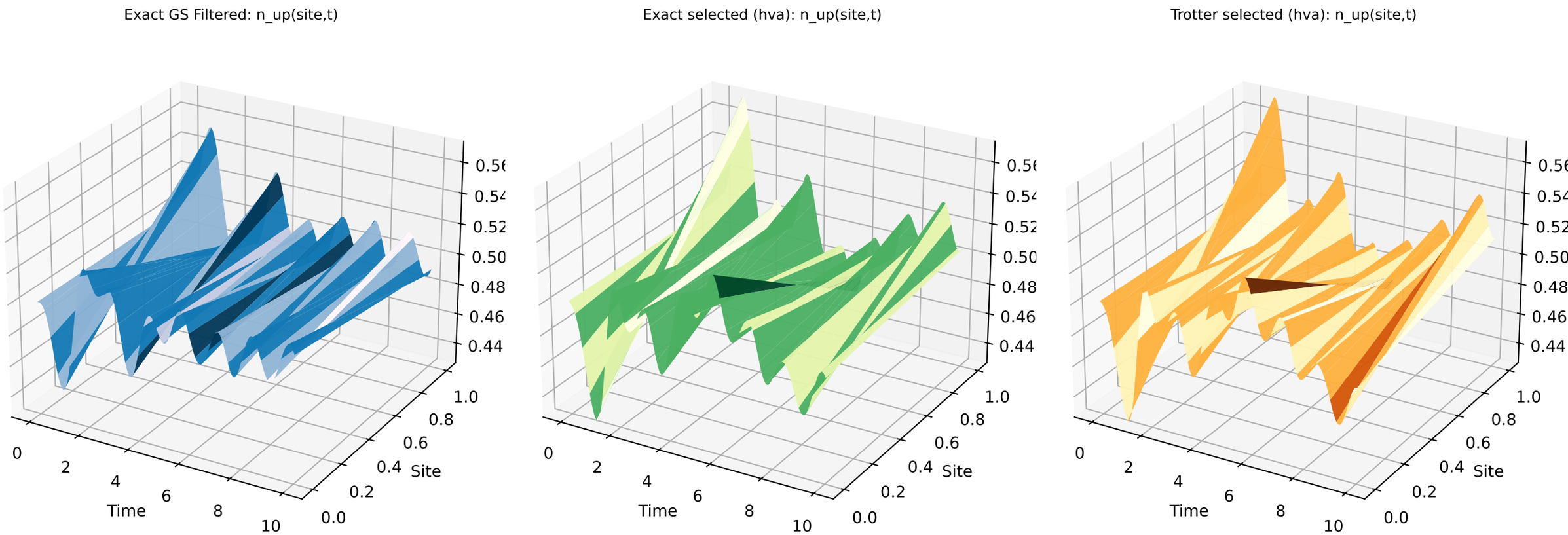
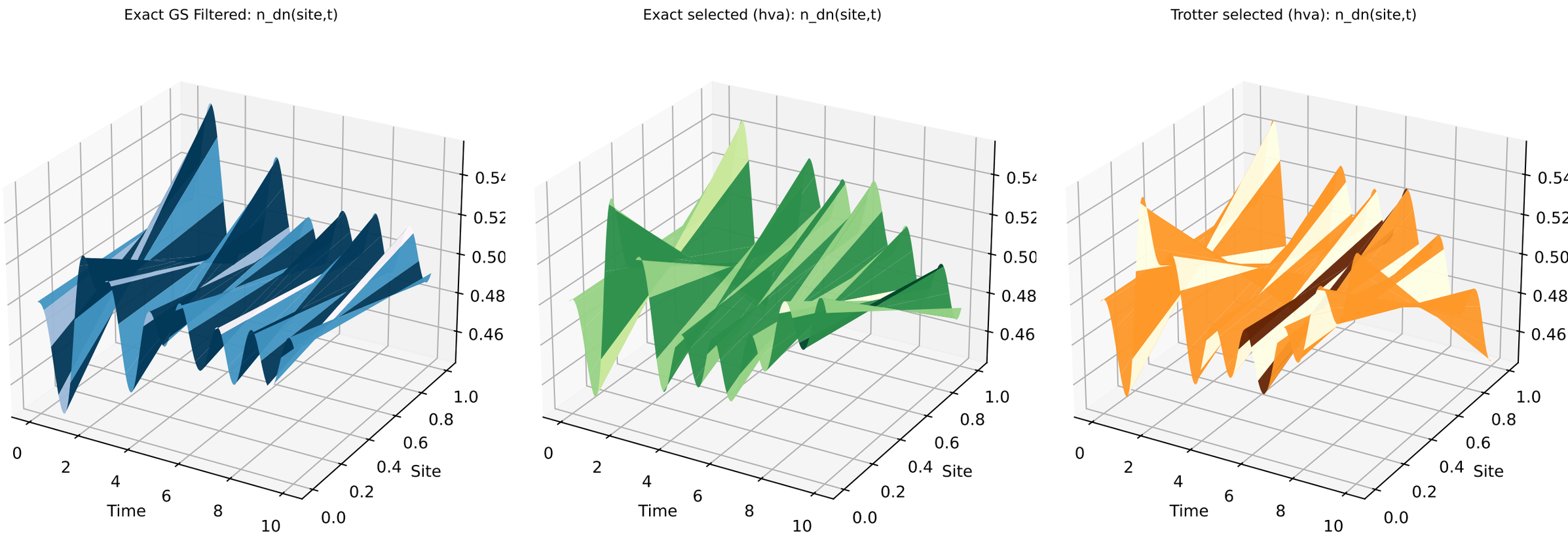

L=2 3D Densities (Total n)



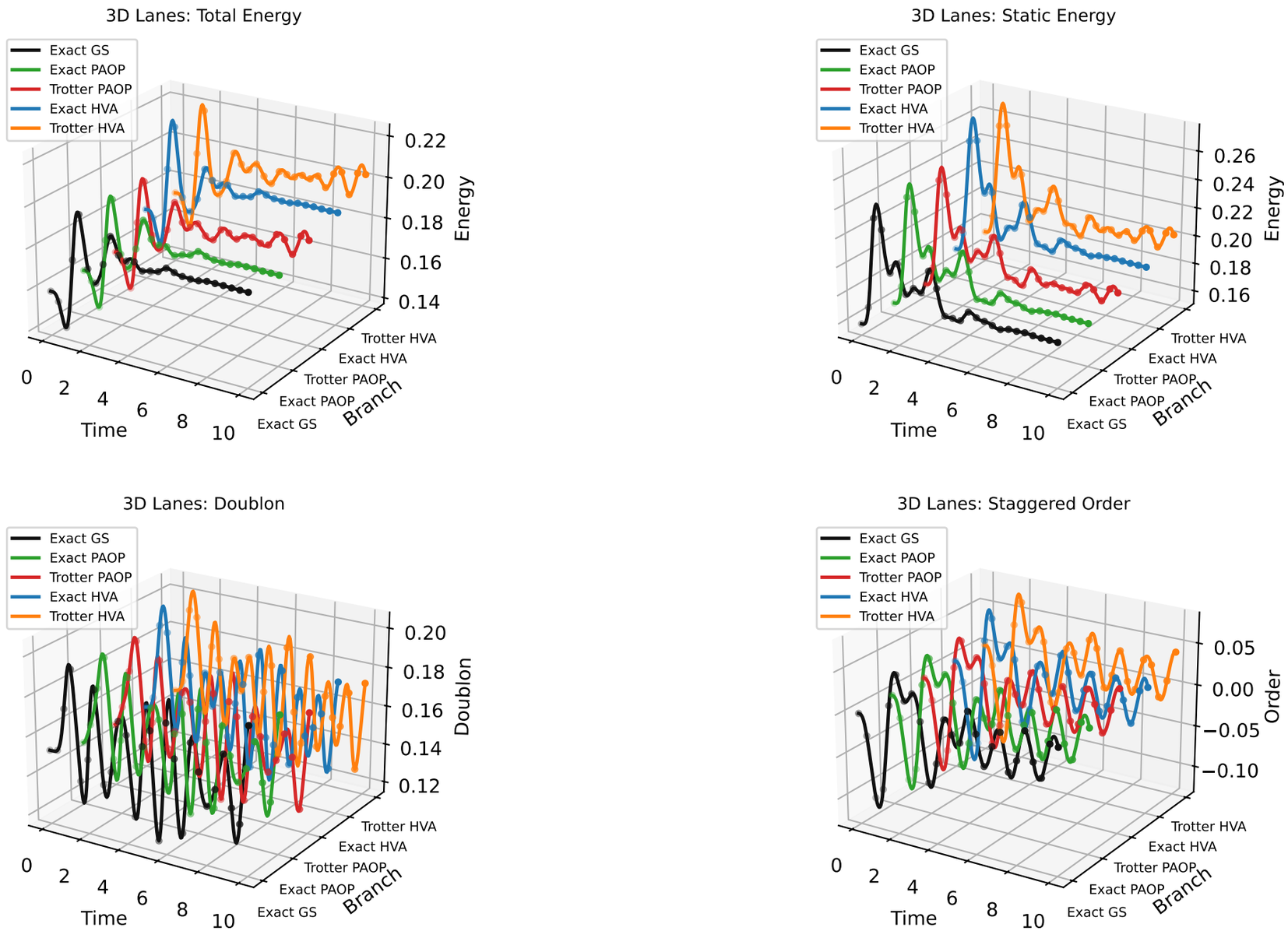
L=2 3D Densities (Spin-Up)

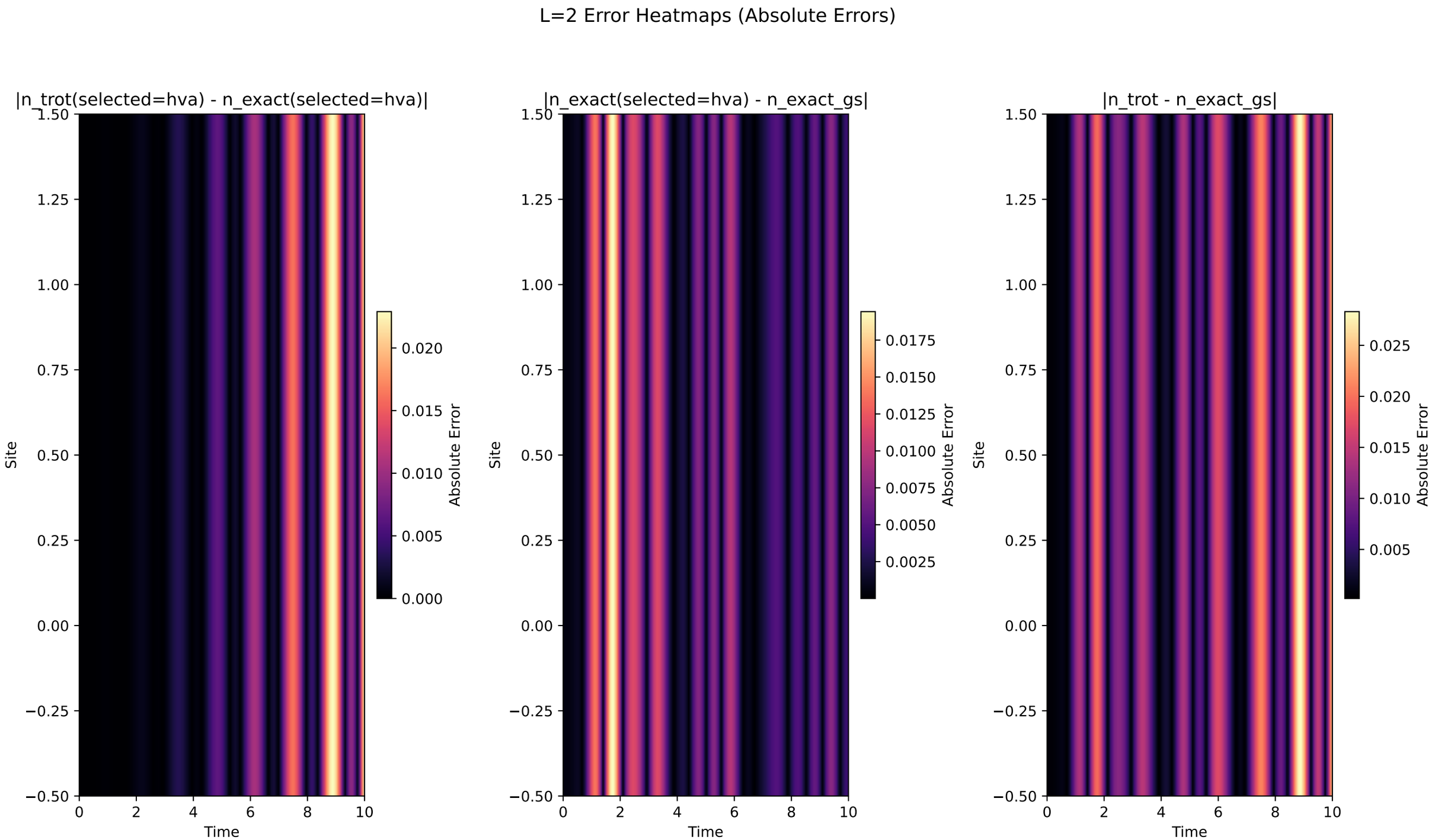


L=2 3D Densities (Spin-Down)

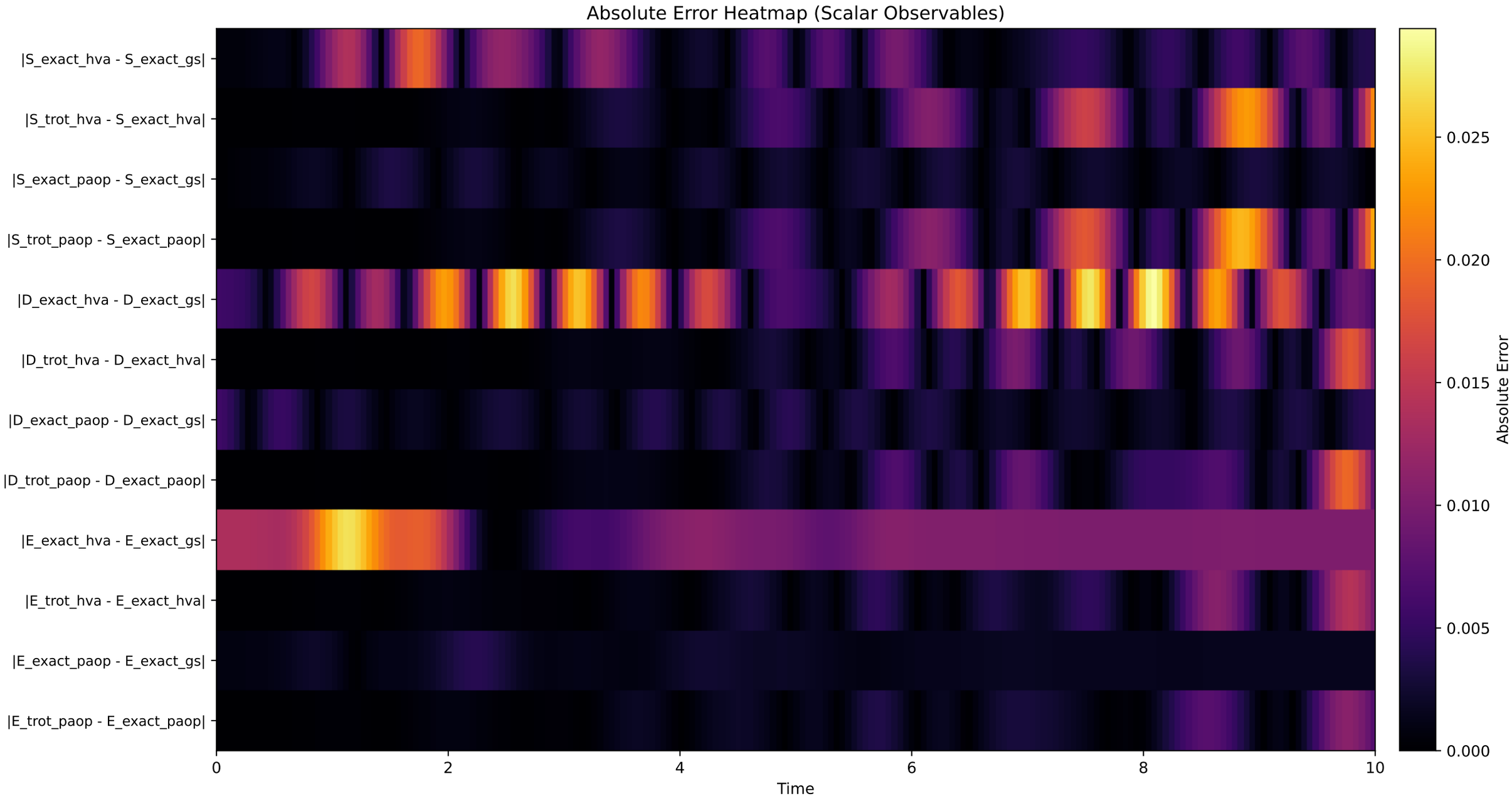


L=2 3D Scalar Observables (Five Branches)

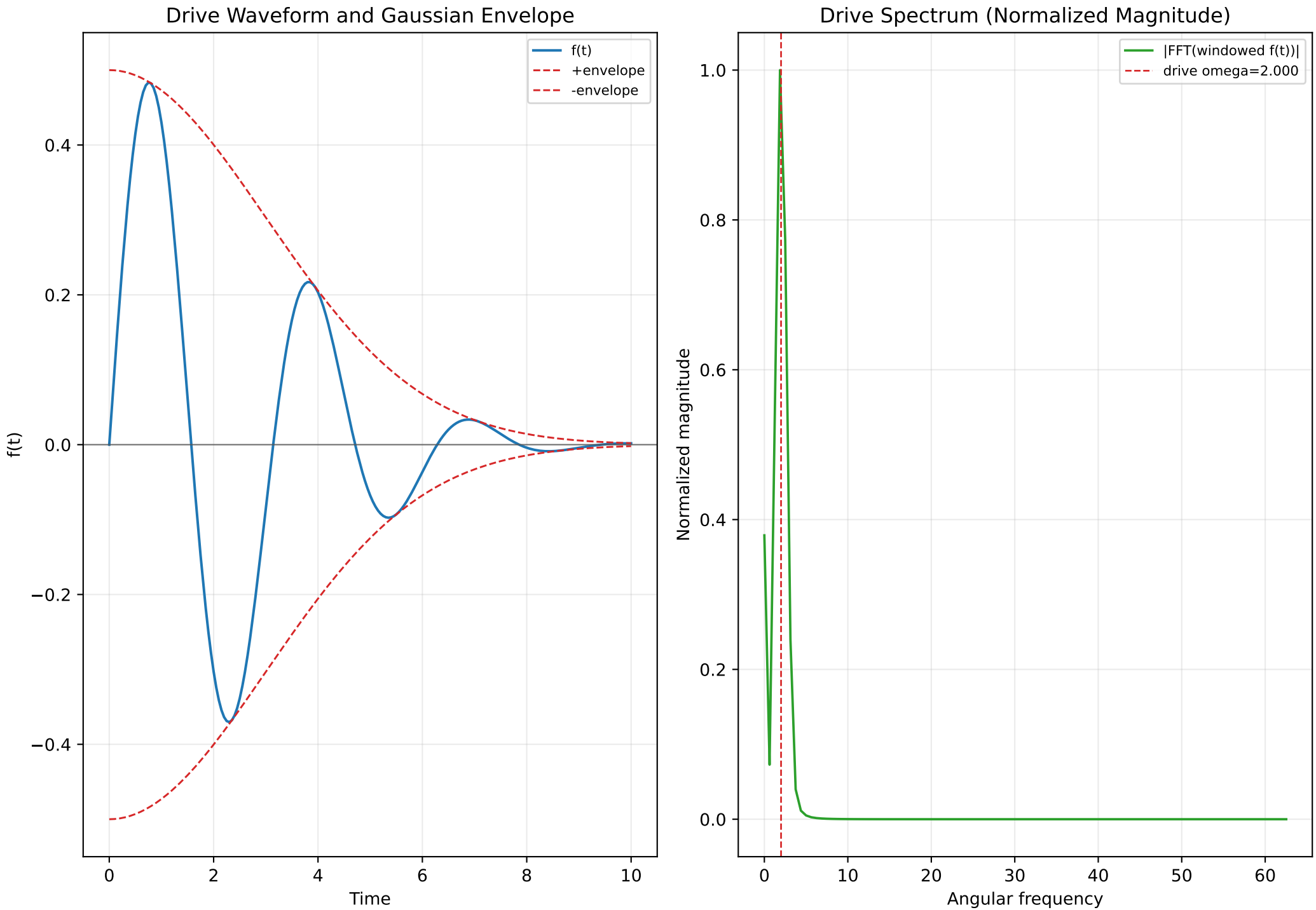



L=2 Error Heatmaps (Absolute Errors)
|n_trot(selected=hva) - n_exact(selected=hva)|
|n_exact(selected=hva) - n_exact_gs|
|n_trot - n_exact_gs|
Site
Time
Absolute Error
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
0
2
4
6
8
10
0.020
0.015
0.010
0.005
0.000
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.025

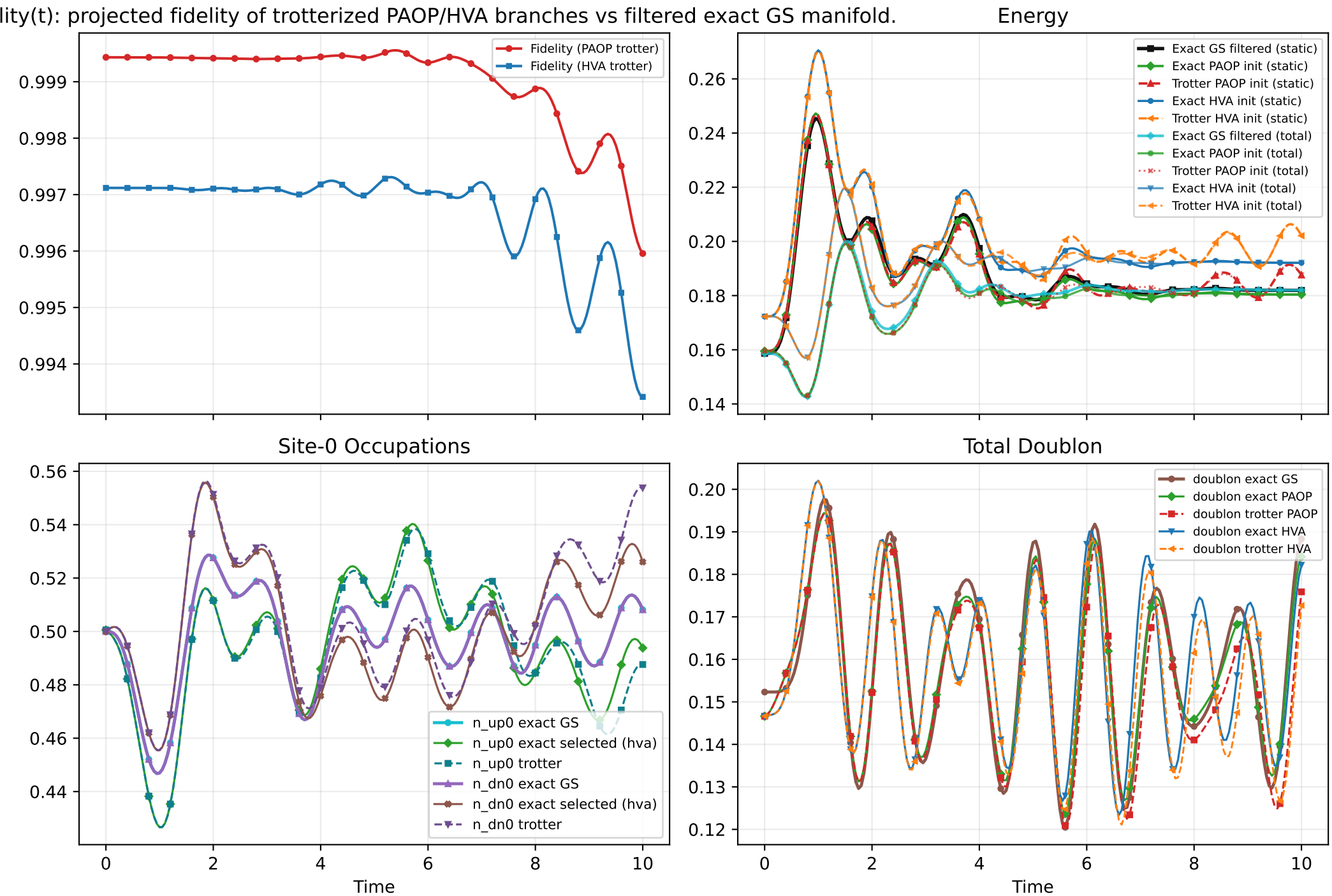
L=2 Scalar Error Heatmap

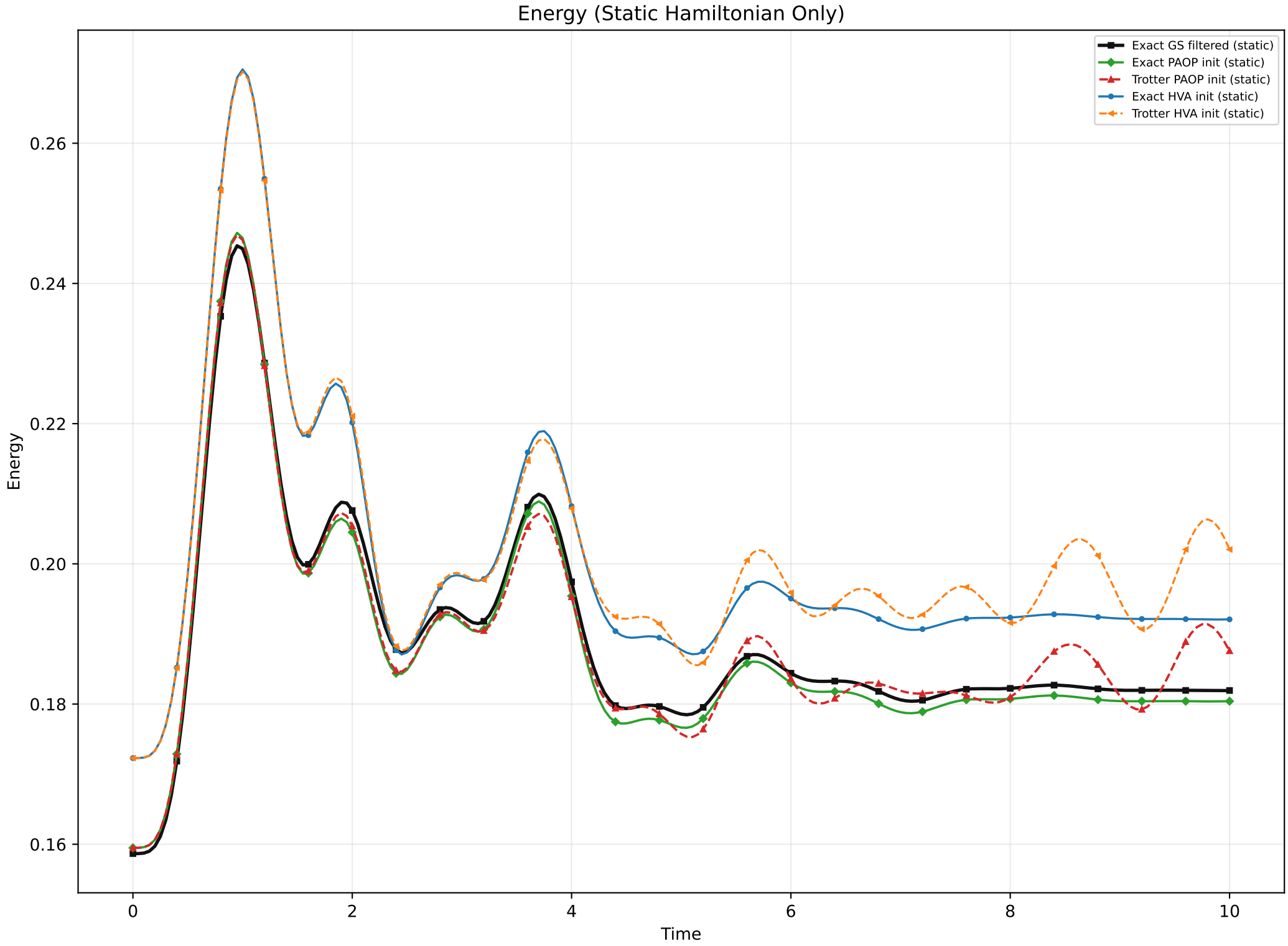


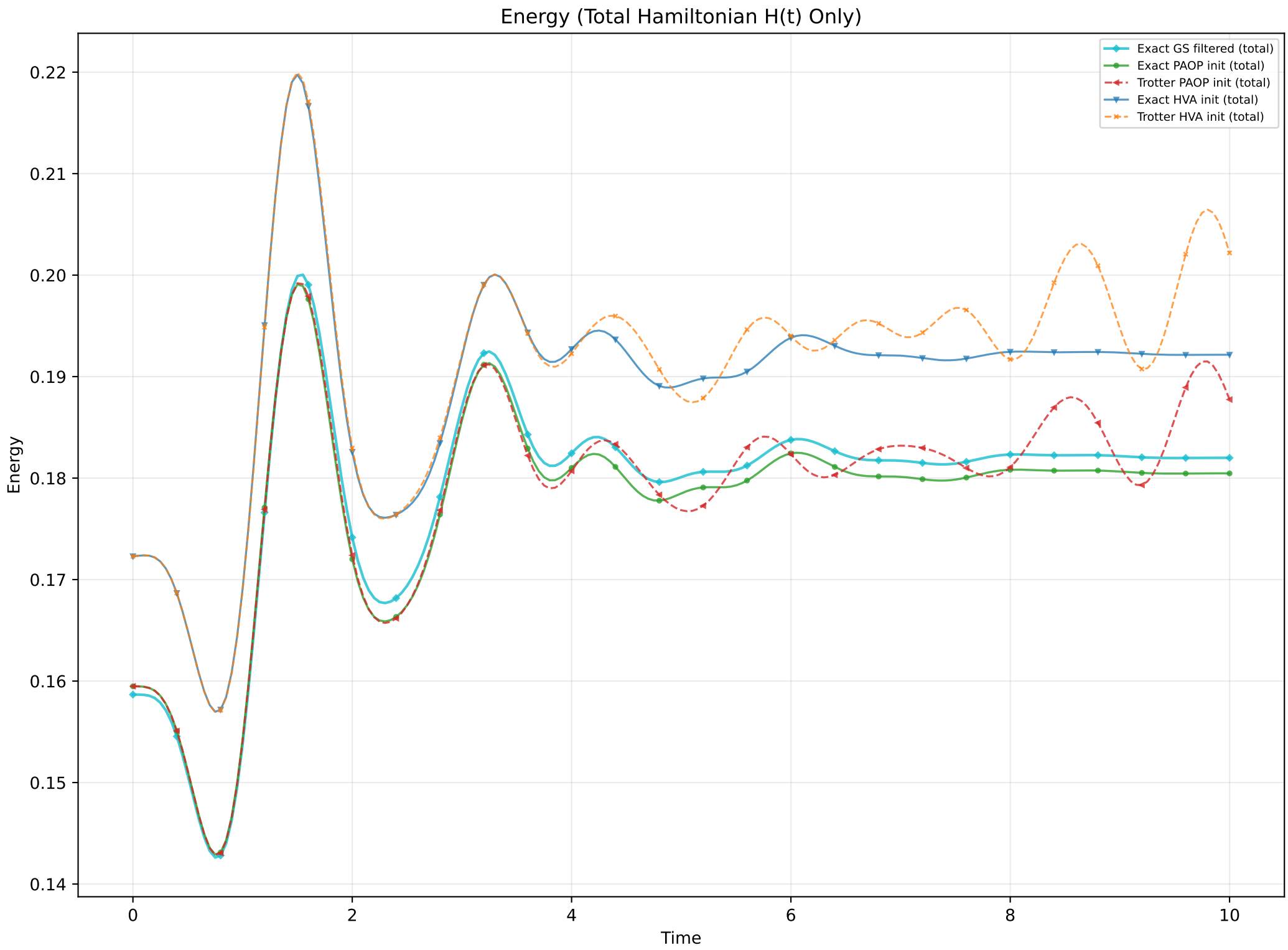
Drive Diagnostics: A=0.5, omega=2.0, tbar=3.0, phi=0.0, t0=0.0



Hardcoded Hubbard Pipeline: L=2



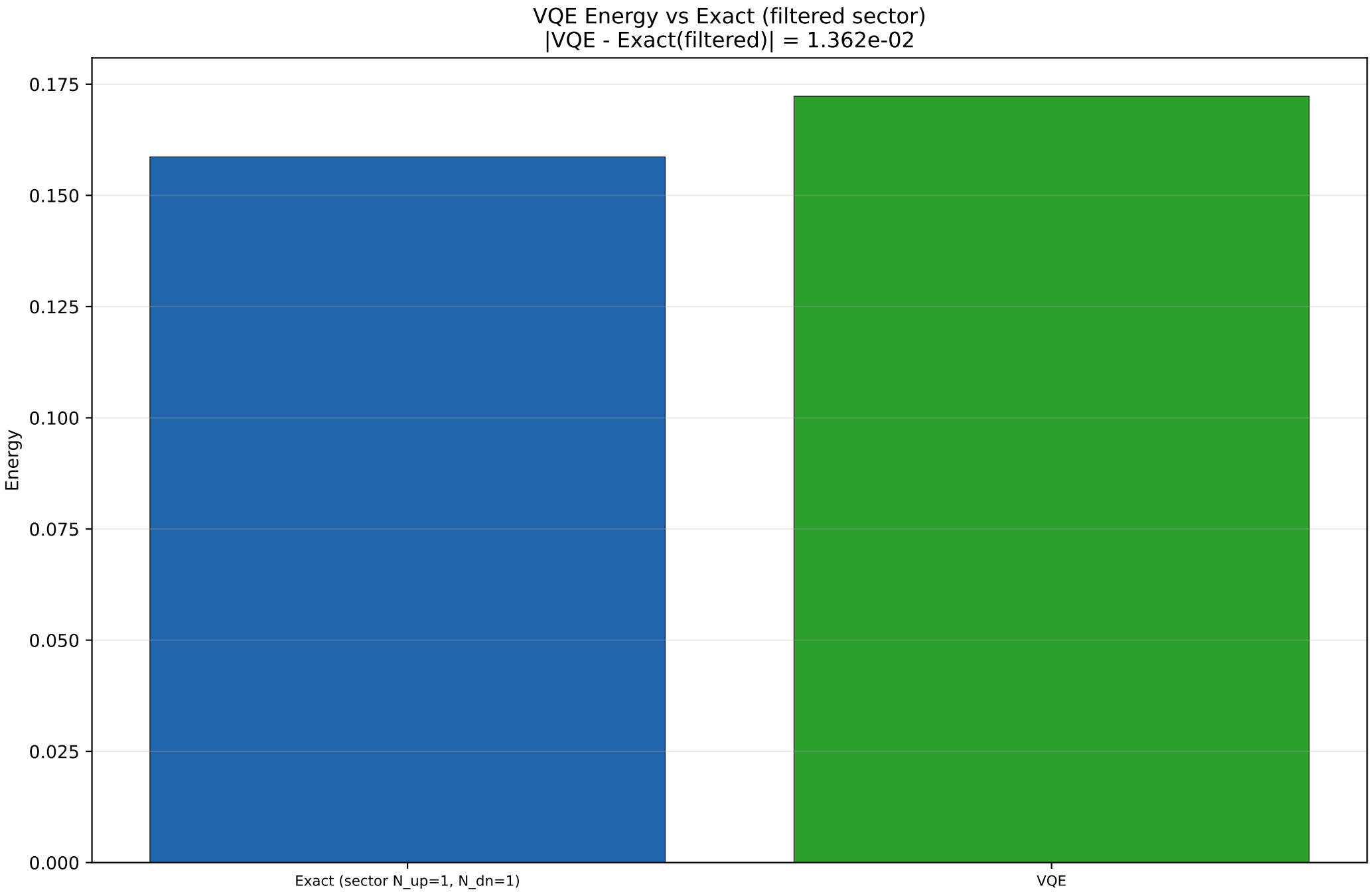

Energy (Static Hamiltonian Only)
Exact GS filtered (static)
Exact PAOP init (static)
Trotter PAOP init (static)
Exact HVA init (static)
Trotter HVA init (static)
Energy
0.26
0.24
0.22
0.20
0.18
0.16
Time
0
2
4
6
8
10


Energy (Total Hamiltonian H(t) Only)
Exact GS filtered (total)
Exact PAOP init (total)
Trotter PAOP init (total)
Exact HVA init (total)
Trotter HVA init (total)
Energy
Time
0.22
0.21
0.20
0.19
0.18
0.17
0.16
0.15
0.14
0
2
4
6
8
10

Appendix — L=2

This section contains supporting/redundant views and metadata.
Core non-redundant analysis appears earlier in the PDF.

VQE optimises within the half-filled sector; exact (filtered) is the true sector ground state.
Full-Hilbert exact energy is in the JSON text summary only.



```
Hardcoded Hubbard pipeline summary

Ansatz:
  - hardcoded ansatz: hh_hva_ptw
  - vqe method: hardcoded_hh_hva_physical_termwise_statevector

Energy + Fidelity:
  - subspace_fidelity_at_t0: 0.9971190883301202
  - subspace_fidelity_paop_t0: 0.999433158001222
  - subspace_fidelity_hva_t0: 0.9971190883301202
  - energy_t0_exact_gs: 0.15866790412572584
  - energy_t0_exact_paop: 0.15948433566498801
  - energy_t0_trotter_paop: 0.159484335664988
  - energy_t0_exact_hva: 0.172290428451842
  - energy_t0_trotter_hva: 0.172290428451842
  - ground_state_exact_energy_full_hilbert: -0.150680123886
  - ground_state_exact_energy_filtered: 0.15866790412572704
  - filtered_sector: {'n_up': 1, 'n_dn': 1}
  - vqe_energy: 0.17229042845183531
  - qpe_energy_estimate: None

Config:
  - initial_state_source: vqe
  - exact_trajectory_label: Exact_Hardcode
  - exact_trajectory_method: python_matrix_eigendecomposition
  - fidelity_definition: fidelity_paop_trotter(t) = <psi_paop_trot(t)|P_exact_gs_subspace(t)|psi_paop_trot(t)> and
    fidelity_hva_trotter(t) = <psi_hva_trot(t)|P_exact_gs_subspace(t)|psi_hva_trot(t)>, where
    P_exact_gs_subspace(t) projects onto the time-evolved filtered-sector ground manifold selected by E <= E0 +
    tol. Legacy key fidelity(t) follows the selected initial-state branch.
  - hamiltonian_terms: 17
  - reference_sanity: {'checked': False, 'reason': 'no matching bundled reference for these settings'}
```